	SAMRAT ASHOK TECHNOLOGICAL INSTITUTE (DEGREE) VIDISHA (M.P)
Department	Applied Science
Branch	Common for All Branches
Session	Batch 2026-27 onwards

Category of Course	Course Title	Course Code	Credits - 3			Theory Paper
			L	T	P	
BSC	Statistics: Probability Distributions and Differential Equations	MA26102	3	0	0	Max.Marks - 70 Min.Marks - 22 Duration - 3 Hrs.

Course Code	Course Title	Maximum Marks Allotted						Total Marks
		Theory Paper			Practical			
		End Sem.	Mid Sem. MST	Quiz/ Assignment	End Sem.	Lab Work	Assign/Quiz /Term paper	
MA26102	Statistics: Probability Distributions and Differential Equations	70	20	10	--	--	--	100

Course Objectives	The objective of this course is to familiarize the prospective engineers with techniques in Differential equations and Statistics. It aims to equip the students with standard concepts and tools at an intermediate to advanced level that will serve them well towards tackling more advanced level of mathematics and applications that they would find useful in their disciplines.
Prerequisite Knowledge	Basics of Differentiations, Integrations and Statistics.
Course Description	This course introduces the fundamentals of probability distributions, sampling theory, and ordinary and partial differential equations. It equips students with mathematical techniques for modelling, analysing, and solving engineering and scientific problems involving uncertainty and dynamic systems.
Course Outcomes	**Upon completion of the course, student will be able to** 1. Acquire the basic knowledge of Statistics: Probability Distributions with their applications and fitting of curves using method of least squares. 2. Learn the principal concepts about sampling and its advantages and also categorized the sampling methods. 3. The Effective Mathematical Tools for the Solutions of Differential Equations that Model Physical Processes. 4. Differential Equation for Solving Engineering Problems **5.** Partial Differential Equations are very much useful for Solving Various Boundary Value Problems

Syllabus

Unit - I	Probability Distribution I : Binomial, Poisson and Normal distributions and their Mean and Variance, Methods of Least Squares and curve fitting.
Unit - II	Probability Sampling distributions: t, F, χ^2 distributions and their applications.
Unit – III	Differential Equations: Differential Equations of first order and first degree, first order and higher degree, Linear Differential Equations, Non-linear Differential Equations, Linear Differential of Higher orders with constant coefficient.
Unit - IV	Differential Equation of other Types: Homogeneous Linear Differential Equations, Legendre Linear Equation, Simultaneous Linear Differential Equation. Method of Variation of Parameters.

Unit – V	Partial Differential Equations: Definition and formation of Partial Differential Equations, Lagrange's Linear PDE, Non-linear PDE, Linear Partial Differential Equation of Second Order with Constant Coefficients. Applications of PDE (Wave equation and Heat Equations)

Text Books

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi. (Indian Author)
2. B.V. Ramana, Higher Engineering Mathematics, McGraw Hill Education, India. (Indian Author)
3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley India.
4. T. Veerarajan, Probability, Statistics and Random Processes, 4th Edition, McGraw Hill Education.

Reference Books

1. R.K. Jain & S.R.K. Iyengar, Advanced Engineering Mathematics, Narosa Publishing House. (Indian Authors)
2. H.K. Dass & Er. Rajnish Verma, Higher Engineering Mathematics, S. Chand Publishing. (Indian Authors)
3. William E. Boyce & Richard C. DiPrima, Elementary Differential Equations and Boundary Value Problems, 11th Edition, Wiley.
4. Sheldon M. Ross, A First Course in Probability, 10th Edition, Pearson Education.